

Question	Marking guidance	Mark	Comments
08.1	Accept any three suitable properties e.g.: • Is a metabolite • Is a solvent • Has a (relatively) high heat capacity • Has a (relatively) large latent heat of vaporisation / evaporation • Has cohesion / hydrogen bonds between molecules;;;	3 max	No explanations are needed However do not accept 'polar' unqualified
08.2	Dilution series;	1	Accept serial dilution
08.3	1. Axes correct way round with linear scales; 2. Axes labelled with mol dm^{-3} and ratio without units; 3. Correct values correctly plotted and suitable curve drawn;	3	3. Accept point to point or smooth curve but no extrapolation
08.4	1. (0.8 mol dm^{-3} sucrose) solution has a <u>more</u> negative / <u>lower</u> water potential than potato (cytoplasm); OR potato (cytoplasm) has a <u>less</u> negative / <u>higher</u> water potential than (0.8 mol dm^{-3} sucrose) solution; 2. (therefore) water moves out (of potato) into the (sucrose) solution by <u>osmosis</u> (so cells decrease in mass);	2	1. Accept sucrose solution is hypertonic / potato cytoplasm is hypotonic 2. Accept water moves down a water potential gradient